TEMA 2 - INSTALACIÓN, CONFIGURACIÓN Y DOCUMENTACIÓN DEL ENTORNO DE DESARROLLO Y DEL ENTORNO DE EXPLOTACIÓN

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EJERCICIOS

1. Instalación y configuración del entorno de desarrollo para la asignatura de Desarrollo web del lado servidor sobre dos máquinas virtuales XXX-USED y XXX-WXED. Documentando la configuración elegida y los aspectos más importantes del proceso de instalación. Documentar el proceso de monitorización del correcto funcionamiento de los servicios. Mantener actualizada esta documentación durante todo el curso.

BNR-USED

1. Configuración inicial.

- Configuración ip.

```
miadmin@bnr-used:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:c8:02:d2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.3.207/24 brd 192.168.3.255 scope global enp0s3
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fec8:2d2/64 scope link
       valid_lft forever preferred_lft forever
miadmin@bnr-used:~$ netstat -r -n
Kernel IP routing table
Destination     Gateway         Genmask         Flags   MSS Window  irtt Iface
0.0.0.0         192.168.3.1     0.0.0.0         UG        0 0          0 enp0s3
192.168.3.0     0.0.0.0         255.255.255.0   U         0 0          0 enp0s3
```

```
# This is the network config written by 'subiquity'
network:
  ethernets:
    enp0s3:
      addresses:
      - 192.168.3.207/24
      dhcp4: false
      routes:
        - to: default
          via: 192.168.3.1
      nameservers:
        addresses:
          - 8.8.8.8
          - 8.8.4.4
  version: 2
```

- Conectividad a Internet.

```
miadmin@bnr-used:~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=114 time=12.9 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=114 time=23.5 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=114 time=13.3 ms
^C
--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2289ms
rtt min/avg/max/mdev = 12.915/16.554/23.497/4.911 ms
miadmin@bnr-used:~$ ping www.google.es
PING www.google.es (142.250.200.131) 56(84) bytes of data.
64 bytes from mad41s14-in-f3.1e100.net (142.250.200.131): icmp_seq=1 ttl=114 time=10.8 ms
64 bytes from mad41s14-in-f3.1e100.net (142.250.200.131): icmp_seq=2 ttl=114 time=11.2 ms
64 bytes from mad41s14-in-f3.1e100.net (142.250.200.131): icmp_seq=3 ttl=114 time=11.0 ms
^C
--- www.google.es ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2104ms
rtt min/avg/max/mdev = 10.759/10.960/11.150/0.159 ms
```

- Nombre de la máquina.

```
miadmin@bnr-used:~$ hostnamectl
 Static hostname: bnr-used
       Icon name: computer-vm
         Chassis: vm
      Machine ID: 796959f77c9145d78e5389ed2dcc4291
         Boot ID: 0eacbf4bf6ae448ba2d420de51d188d6
  Virtualization: oracle
Operating System: Ubuntu 22.04.3 LTS
          Kernel: Linux 5.15.0-84-generic
    Architecture: x86-64
 Hardware Vendor: innotek GmbH
  Hardware Model: VirtualBox
```

- Usuario inicial miadmin.

```
miadmin@bnr-used:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
_apt:x:100:65534::/nonexistent:/usr/sbin/nologin
systemd-network:x:101:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:102:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:103:104::/nonexistent:/usr/sbin/nologin
systemd-timesync:x:104:105:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
pollinate:x:105:1::/var/cache/pollinate:/bin/false
sshd:x:106:65534::/run/sshd:/usr/sbin/nologin
syslog:x:107:113::/home/syslog:/usr/sbin/nologin
uuidd:x:108:114::/run/uuidd:/usr/sbin/nologin
tcpdump:x:109:115::/nonexistent:/usr/sbin/nologin
tss:x:110:116:TPM software stack,,,:/var/lib/tpm:/bin/false
landscape:x:111:117::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:112:118:fwupd-refresh user,,,:/run/systemd:/usr/sbin/nologin
usbmux:x:113:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
miadmin:x:1000:1000:bn-used:/home/miadmin:/bin/bash
lxd:x:999:100::/var/snap/lxd/common/lxd:/bin/false
```

- Activar el cortafuegos y abrir el puerto 22 para la conexión por ssh.

```
miadmin@bnr-used:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
miadmin@bnr-used:~$ sudo ufw allow 22
Rule updated
Rule updated (v6)
miadmin@bnr-used:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22                         ALLOW       Anywhere
22 (v6)                    ALLOW       Anywhere (v6)
```

2. Apache

- Instalación del servicio apache.

```
miadmin@bnr-used:~$ sudo apt-get update
[sudo] password for miadmin:
Obj:1 http://es.archive.ubuntu.com/ubuntu jammy InRelease
Obj:2 http://es.archive.ubuntu.com/ubuntu jammy-updates InRelease
Obj:3 http://es.archive.ubuntu.com/ubuntu jammy-backports InRelease
Obj:4 http://es.archive.ubuntu.com/ubuntu jammy-security InRelease
Leyendo lista de paquetes... Hecho
miadmin@bnr-used:~$ sudo apt install apache2
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
Se instalarán los siguientes paquetes adicionales:
```

- Comprobar el servicio.

```
miadmin@bnr-used:~$ sudo service apache2 status
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
     Active: active (running) since Wed 2023-10-04 08:10:53 UTC; 1min 30s ago
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 2062 (apache2)
      Tasks: 55 (limit: 2220)
     Memory: 5.0M
        CPU: 35ms
     CGroup: /system.slice/apache2.service
             ├─2062 /usr/sbin/apache2 -k start
             ├─2064 /usr/sbin/apache2 -k start
             └─2065 /usr/sbin/apache2 -k start

oct 04 08:10:53 bnr-used systemd[1]: Starting The Apache HTTP Server...
oct 04 08:10:53 bnr-used apachectl[2061]: AH00558: apache2: Could not reliably determine the server's
oct 04 08:10:53 bnr-used systemd[1]: Started The Apache HTTP Server.
lines 1-16/16 (END)
```

- Abrimos el puerto de apache en el cortafuegos.

```
miadmin@bnr-used:~$ sudo ufw allow Apache
[sudo] password for miadmin:
Rule added
Rule added (v6)
miadmin@bnr-used:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22                         ALLOW       Anywhere
Apache                     ALLOW       Anywhere
22 (v6)                    ALLOW       Anywhere (v6)
Apache (v6)                ALLOW       Anywhere (v6)
```

- Creamos el usuario con su contraseña y directorio y lo metemos en el grupo www-data.

```
miadmin@bnr-used:~$ sudo useradd -d /var/www/html/ operadorweb
miadmin@bnr-used:~$ sudo passwd operadorweb
New password:
Retype new password:
passwd: password updated successfully
miadmin@bnr-used:~$ sudo usermod -G www-data operadorweb
miadmin@bnr-used:~$ cat /etc/passwd | grep operadorweb
operadorweb:x:1001:1001::/var/www/html/:/bin/sh
miadmin@bnr-used:~$ cat /etc/group
group   group-
miadmin@bnr-used:~$ cat /etc/group | grep www-data
www-data:x:33:operadorweb
```

- Ponemos que el usuario operador web sea el propietario del directorio y cambiamos los permisos del mismo.

```
miadmin@bnr-used:~$ sudo chown --recursive operadorweb:www-data /var/www/html
miadmin@bnr-used:~$ sudo chmod -R 2775 /var/www/html
miadmin@bnr-used:~$ ls -l /var/www
total 4
drwxrwsr-x 2 operadorweb www-data 4096 oct  4 08:10 html
```

- Cambiamos esa directiva en el fichero de configuración de apache para el .htaccess.

```
miadmin@bnr-used: ~
  GNU nano 6.2                                          /etc/apache2/apache2.conf *
<Directory />
        Options FollowSymLinks
        AllowOverride None
        Require all denied
</Directory>

<Directory /usr/share>
        AllowOverride None
        Require all granted
</Directory>

<Directory /var/www/>
        Options Indexes FollowSymLinks
        AllowOverride All
        Require all granted
</Directory>
```

- Entramos en el usuario operador web y dentro del directorio de este usuario(var/www/html) creamos un. htacces y escribimos lo que se ve en la captura.

```
miadmin@bnr-used:/var/www/html$ su operadorweb
Password:
  GNU nano 6.2
DirectoryIndex index.php index.html
```

3. PHP

- Actualizamos los repositorios y instalamos la versión 8.1 de php.

```
miadmin@bnr-used:~$ sudo apt update

miadmin@bnr-used:~$ sudo apt -y install php8.1
```

- Instalamos la siguiente librería para php.

```
miadmin@bnr-used:~$ sudo apt install libapache2-mod-php
```

- Instalamos los mods siguientes:

```
miadmin@bnr-used:~$ sudo apt install -y php8.1-xml

miadmin@bnr-used:~$ sudo apt install -y php8.1-soap
```

- Editamos el fichero php.ini.

```
miadmin@bnr-used: ~
  GNU nano 6.2                                                    /etc/php/8.1/apache2/php.ini *
;                     of PHP
; E_USER_DEPRECATED - user-generated deprecation warnings
;
; Common Values:
;   E_ALL (Show all errors, warnings and notices including coding standards.)
;   E_ALL & ~E_NOTICE  (Show all errors, except for notices)
;   E_ALL & ~E_NOTICE & ~E_STRICT  (Show all errors, except for notices and coding standards warnings.)
;   E_COMPILE_ERROR|E_RECOVERABLE_ERROR|E_ERROR|E_CORE_ERROR  (Show only errors)
; Default Value: E_ALL
; Development Value: E_ALL
; Production Value: E_ALL & ~E_DEPRECATED & ~E_STRICT
; https://php.net/error-reporting
error_reporting = E_ALL & ~E_DEPRECATED & ~E_STRICT

; This directive controls whether or not and where PHP will output errors,
; notices and warnings too. Error output is very useful during development, but
; it could be very dangerous in production environments. Depending on the code
; which is triggering the error, sensitive information could potentially leak
; out of your application such as database usernames and passwords or worse.
; For production environments, we recommend logging errors rather than
; sending them to STDOUT.
; Possible Values:
;   Off = Do not display any errors
;   stderr = Display errors to STDERR (affects only CGI/CLI binaries!)
;   On or stdout = Display errors to STDOUT
; Default Value: On
; Development Value: On
; Production Value: Off
; https://php.net/display-errors
display_errors = On
```

- Reiniciamos servicio.

```
miadmin@bnr-used:~$ sudo service apache2 restart
```

- Ponemos el límite de memoria en 256MB.

```
  GNU nano 6.2                                                    /etc/php/8.1/apache2/php.ini
; https://php.net/max-execution-time
; Note: This directive is hardcoded to 0 for the CLI SAPI
max_execution_time = 30

; Maximum amount of time each script may spend parsing request data. It's a good
; idea to limit this time on productions servers in order to eliminate unexpectedly
; long running scripts.
; Note: This directive is hardcoded to -1 for the CLI SAPI
; Default Value: -1 (Unlimited)
; Development Value: 60 (60 seconds)
; Production Value: 60 (60 seconds)
; https://php.net/max-input-time
max_input_time = 60

; Maximum input variable nesting level
; https://php.net/max-input-nesting-level
;max_input_nesting_level = 64

; How many GET/POST/COOKIE input variables may be accepted
;max_input_vars = 1000

; Maximum amount of memory a script may consume
; https://php.net/memory-limit
memory_limit = 256M
```

- Reiniciamos el servicio otra vez.

```
miadmin@bnr-used:~$ sudo service apache2 restart
```

4.Xdebug.

- Comprobamos que está activo el módulo xdebug.

```
miadmin@bnr-used:~$ php -m  | grep xdebug
miadmin@bnr-used:~$ php -m xdebug
[PHP Modules]
calendar
Core
ctype
date
dom
exif
FFI
fileinfo
filter
ftp
gettext
hash
iconv
json
libxml
openssl
pcntl
pcre
PDO
Phar
posix
readline
Reflection
session
shmop
SimpleXML
soap
sockets
sodium
SPL
standard
sysvmsg
sysvsem
sysvshm
tokenizer
xml
xmlreader
xmlwriter
xsl
Zend OPcache
zlib

[Zend Modules]
Zend OPcache
```

- Actualizamos los repositorios e instalamos el lo que se ve en la siguiente captura.

```
miadmin@bnr-used:~$ sudo apt update
Obj:1 http://es.archive.ubuntu.com/ubuntu jammy InRelease
Des:2 http://es.archive.ubuntu.com/ubuntu jammy-updates InRelease [119 kB]
Des:3 http://es.archive.ubuntu.com/ubuntu jammy-backports InRelease [109 kB]
Des:4 http://es.archive.ubuntu.com/ubuntu jammy-security InRelease [110 kB]
Des:5 http://es.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [1.057 kB]
Des:6 http://es.archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [232 kB]
Des:7 http://es.archive.ubuntu.com/ubuntu jammy-updates/main amd64 c-n-f Metadata [15,6 kB]
Des:8 http://es.archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [969 kB]
Des:9 http://es.archive.ubuntu.com/ubuntu jammy-updates/restricted Translation-en [156 kB]
Des:10 http://es.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [989 kB]
Des:11 http://es.archive.ubuntu.com/ubuntu jammy-updates/universe Translation-en [216 kB]
Des:12 http://es.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 c-n-f Metadata [21,9 kB]
Des:13 http://es.archive.ubuntu.com/ubuntu jammy-security/main amd64 Packages [856 kB]
Des:14 http://es.archive.ubuntu.com/ubuntu jammy-security/main Translation-en [175 kB]
Des:15 http://es.archive.ubuntu.com/ubuntu jammy-security/main amd64 c-n-f Metadata [11,4 kB]
Des:16 http://es.archive.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [953 kB]
Des:17 http://es.archive.ubuntu.com/ubuntu jammy-security/restricted Translation-en [154 kB]
Des:18 http://es.archive.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [788 kB]
Des:19 http://es.archive.ubuntu.com/ubuntu jammy-security/universe Translation-en [144 kB]
Des:20 http://es.archive.ubuntu.com/ubuntu jammy-security/universe amd64 c-n-f Metadata [16,7 kB]
Descargados 7.093 kB en 2s (4.230 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Se pueden actualizar 34 paquetes. Ejecute «apt list --upgradable» para verlos.
miadmin@bnr-used:~$ sudo apt install php8.1-xdebug
```

- Ponemos la siguiente directiva en el fichero de configuración de xdebug.

```
miadmin@bnr-used: ~
  GNU nano 6.2                                    /etc/php/8.1/apache2/conf.d/20-xdebug.ini *
zend_extension=xdebug.so
xdebug.mode=debug
xdebug.client.host=localhost
xdebug.client.port=9003
xdebug.jdekey="netbeans-debug"
```

- Reiniciamos el servicio apache y abrimos el puerto 9003.

```
miadmin@bnr-used:~$ sudo service apache2 restart
```

```
miadmin@bnr-used: ~
miadmin@bnr-used:~$ sudo ufw allow 9003
Rule added
Rule added (v6)
```

5.MySQL

- Instalar MySQL server.

```
miadmin@bnr-used:~$ sudo apt install mysql-server
```

- Configurar las siguientes directivas en el archivo de configuración de mysql.

```
miadmin@bnr-used: ~
  GNU nano 6.2                                                    /etc/mysql/mysql.conf.d/mysqld.cnf *
#
# The MySQL database server configuration file.
#
# One can use all long options that the program supports.
# Run program with --help to get a list of available options and with
# --print-defaults to see which it would actually understand and use.
#
# For explanations see
# http://dev.mysql.com/doc/mysql/en/server-system-variables.html

# Here is entries for some specific programs
# The following values assume you have at least 32M ram

[mysqld]
#
# * Basic Settings
#
user            = mysql
# pid-file      = /var/run/mysqld/mysqld.pid
# socket        = /var/run/mysqld/mysqld.sock
# port          = 3306
# datadir       = /var/lib/mysql


# If MySQL is running as a replication slave, this should be
# changed. Ref https://dev.mysql.com/doc/refman/8.0/en/server-system-variables.html#sysvar_tmpdir
# tmpdir                = /tmp
#
# Instead of skip-networking the default is now to listen only on
# localhost which is more compatible and is not less secure.
#bind-address           = 127.0.0.1
#mysqlx-bind-address    = 127.0.0.1
```

- Reiniciar el servicio.

```
miadmin@bnr-used:~$ sudo service mysql restart
```

- Comprobamos que esta escuchando por el puerto 3306 y lo abrimos.

```
miadmin@bnr-used:~$ ss -punta
Netid               State               Recv-Q              Send-Q                          Local Address:Port                              Peer Address:Port
udp                 UNCONN              0                   0                               127.0.0.53%lo:53                                    0.0.0.0:*
tcp                 LISTEN              0                   4096                            127.0.0.53%lo:53                                    0.0.0.0:*
tcp                 LISTEN              0                   128                                   0.0.0.0:22                                    0.0.0.0:*
tcp                 ESTAB               0                   52                              192.168.3.207:22                                192.168.3.7:53893
tcp                 LISTEN              0                   511                                      *:80                                          *:*
tcp                 LISTEN              0                   128                                   [::]:22                                       [::]:*
tcp                 LISTEN              0                   70                                       *:33060                                       *:*
tcp                 LISTEN              0                   151                                      *:3306                                        *:*
miadmin@bnr-used:~$ sudo ufw allow 3306
Rule added
Rule added (v6)
```

- Ejecutamos el siguiente script. Cuando nos aparezcan diferentes opciones le damos a todo que si y ponemos 0 en una opción de ellas.

```
miadmin@bnr-used:~$ sudo mysql_secure_installation
```

- Instalamos lo siguiente.

```
miadmin@bnr-used:~$ sudo apt install libapache2-mod-php8.1 php8.1-mysql
```

- Creamos este usuario.

```
mysql> CREATE USER 'admindb'@'%' IDENTIFIED BY 'P@ssw0rd';
Query OK, 0 rows affected (0,04 sec)
```

- Le damos todos los privilegios.

```
mysql> GRANT ALL PRIVILEGES ON *.* TO 'admindb'@'%' WITH GRANT OPTION;
Query OK, 0 rows affected (0,01 sec)
```

- Comprobamos conexión con l usuario.

```
miadmin@bnr-used:~$ mysql -u admindb -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 14
Server version: 8.0.34-0ubuntu0.22.04.1 (Ubuntu)

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owners.
```

- Comprobamos posibles errores.

```
miadmin@bnr-used:~$ tail -n200 /var/log/mysql/error.log | grep ERROR
```

- Con este comando limpiamos el archivo de errores

```
miadmin@bnr-used:~$ mysqladmin -u admindb -p flush-logs error
```

- Cambiamos los requisitos de la contraseña para poder poner de contraseña paso en vez de P@sw0rd.

```
mysql> SHOW VARIABLES LIKE 'validate_password%';
+-----------------------------------------------+--------+
| Variable_name                                 | Value  |
+-----------------------------------------------+--------+
| validate_password.changed_characters_percentage | 0      |
| validate_password.check_user_name             | ON     |
| validate_password.dictionary_file             |        |
| validate_password.length                      | 8      |
| validate_password.mixed_case_count            | 1      |
| validate_password.number_count                | 1      |
| validate_password.policy                      | MEDIUM |
| validate_password.special_char_count          | 1      |
+-----------------------------------------------+--------+
8 rows in set (0,02 sec)

mysql> SET GLOBAL validate_password.length = 4;
Query OK, 0 rows affected (0,00 sec)

mysql> SET GLOBAL validate_password.policy = LOW;
Query OK, 0 rows affected (0,00 sec)
```

- Creamos el usuario siguiendo la guía de mysql.

```
mysql> CREATE USER 'adminsql'@'%' IDENTIFIED BY 'paso';
Query OK, 0 rows affected (0,06 sec)

mysql> GRANT ALL PREVILEGES ON *.* TO 'adminsql'@'%' WITH GRANT OPTION;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manu
t line 1
mysql> GRANT ALL PRIVILEGES ON *.* TO 'adminsql'@'%' WITH GRANT OPTION;
Query OK, 0 rows affected (0,00 sec)
```

- Comprobamos que esta instalado entrono al php.

```
miadmin@bnr-used:~$ dpkg --get-selections | grep -i php
libapache2-mod-php                              install
libapache2-mod-php8.1                           install
php-common                                      install
php8.1                                          install
php8.1-cli                                      install
php8.1-common                                   install
php8.1-mysql                                    install
php8.1-opcache                                  install
php8.1-readline                                 install
php8.1-soap                                     install
php8.1-xdebug                                   install
php8.1-xml                                      install
```

- Instalar los siguientes módulos y luego seleccionas apache con el space y en la segunda pantalla le das que si pones paso como contraseña y luego te da error y le das a interrumpir.

```
miadmin@bnr-used:~$ sudo apt install phpmyadmin php-mbstring php-zip php-gd php-json php-curl
```

- Desinstalamos ese componente.

```
mysql> UNINSTALL COMPONENT "file://component_validate_password";
Query OK, 0 rows affected (0,01 sec)
```

- Instalamos phpMyAdmin.

```
miadmin@bnr-used:~$ sudo apt install phpmyadmin
```

- Volvemos a instalar el componente previamente desinstalado, activamos el módulo mbstring y reiniciamos servicio

```
mysql> INSTALL COMPONENT "file://component_validate_password";
Query OK, 0 rows affected (0,03 sec)

mysql> EXIT
Bye
miadmin@bnr-used:~$ sudo phpenmod mbstring
miadmin@bnr-used:~$ sudo systemctl restart apache2
```

BNR-WXED